
Amazon Route 53 – Complete Hands-On Lab Guide

♦ Lab 1: Registering a Domain using Route 53

Goal:

Buy and register a new domain via Route 53.

Steps:

1. Open **Route 53 Console** → Click on “**Registered domains**”.
2. Click “**Register domain**”.
3. Search for a domain (e.g., [myawsdemodomain.com](#)) → Add to cart.
4. Choose **1-year registration**, and proceed.
5. Fill in **contact details**.
6. Complete **email verification** (mandatory).
7. AWS automatically creates a **hosted zone** after domain registration.

Notes:

- AWS becomes your **domain registrar**.
- Takes ~10 minutes to complete.

- You'll manage DNS records inside the created **public hosted zone**.
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♦ Lab 2: Create an A Record in Hosted Zone



Goal:

Point the domain to an EC2 instance or IP using a basic **A Record**.



Steps:

1. Go to **Route 53** → **Hosted Zones** → Click your domain.
2. Click **“Create record”**.
3. Choose:
 - **Record name:** Leave blank for root domain or use **www**.
 - **Record type:** A (IPv4)
 - **Value:** Enter **public IP of EC2**
 - **Routing policy:** Simple
4. Click **Create Record**.



Tip:

- Test by accessing <http://yourdomain.com> in a browser.
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♦ Lab 3: Simple Routing Policy – Demo

✓ Steps:

1. Create one **A Record** for your domain → Point to a single EC2 instance.
2. Choose **Routing policy** as **Simple**.
3. No health check is attached.
4. Open browser → test DNS resolution.

 Use Case: Single web server with no failover.

♦ Lab 4: Weighted Routing Policy – Demo

Goal:

Split traffic between two EC2 instances.

✓ Steps:

1. Launch **2 EC2 instances** in different AZs with Apache or Nginx.
2. In **Hosted Zone**, create:
 - **Record A1:** `mydomain.com` → IP1 → Weight: 70
 - **Record A2:** `mydomain.com` → IP2 → Weight: 30
3. Test by refreshing the browser or using `dig +short yourdomain.com`.

 Tip: Weights control traffic proportion. Add health checks if needed.

♦ Lab 5: Latency Routing Policy – Demo

Goal:

Route traffic to the lowest-latency region.

Steps:

1. Launch EC2 in **us-east-1** and **ap-south-1**.
2. Create **A Records**:
 - One pointing to US EC2 with region set to **US East**
 - Another to Mumbai EC2 with region set to **Asia Pacific**
3. Routing policy: **Latency-based**.
4. Test from **VPN or proxies** in different regions.

 Used for **global websites** to minimize user latency.

♦ Lab 6: Failover Routing Policy – Demo

Goal:

Create a **primary-secondary failover** for web hosting.

Steps:

1. Launch 2 EC2 instances:

- **Primary:** Region A (with index.html saying “Primary”)
 - **Secondary:** Region B (with “Secondary”)
2. Create health check:
- HTTP → Port 80 → Path `/index.html`
3. Create 2 records:
- **Primary Record A:** `mydomain.com` → IP1 → Primary → Attach health check.
 - **Secondary Record A:** `mydomain.com` → IP2 → Secondary.
4. Stop web server on primary → See failover in action.

🧠 Great for **high availability** setup.

♦ Lab 7: Multivalue Answer Routing Policy – Demo



Goal:

Use Route 53 to simulate basic load balancing.



Steps:

1. Launch 3 EC2 instances.
2. Create 3 **A Records** with same domain:
 - Routing policy: **Multivalue**

- Health checks: Optional (can limit to healthy responses)

Test with:

dig +short mydomain.com

- 3.
4. Refresh browser → Multiple responses will be served round-robin.

⚠ Max 8 records allowed with multivalue.

◆ Lab 8: 3rd-Party Domain Integration with Route 53 + Cleanup

Goal:

Use Route 53 as DNS for a domain registered outside AWS.

✓ Steps:

1. Buy a domain from **GoDaddy/Namecheap/etc.**
2. In Route 53, create a **Hosted Zone** with the domain name.
3. Copy **4 Name Servers (NS)** from hosted zone.
4. Go to third-party registrar → Replace domain NS with AWS Route 53 NS.
5. Propagate DNS (~1–24 hrs).
6. Create **A/CNAME/Other records** in hosted zone.

Cleanup:

- Delete hosted zone (after deregistering).
 - Release EC2, terminate health checks.
 - Remove domain if not needed (note: refund not possible on domain).
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Optional Enhancements:

- Enable **DNSSEC** for domain validation.
 - Use **Alias Records** for S3 static website hosting.
 - Integrate **CloudFront** with your domain via Route 53.
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